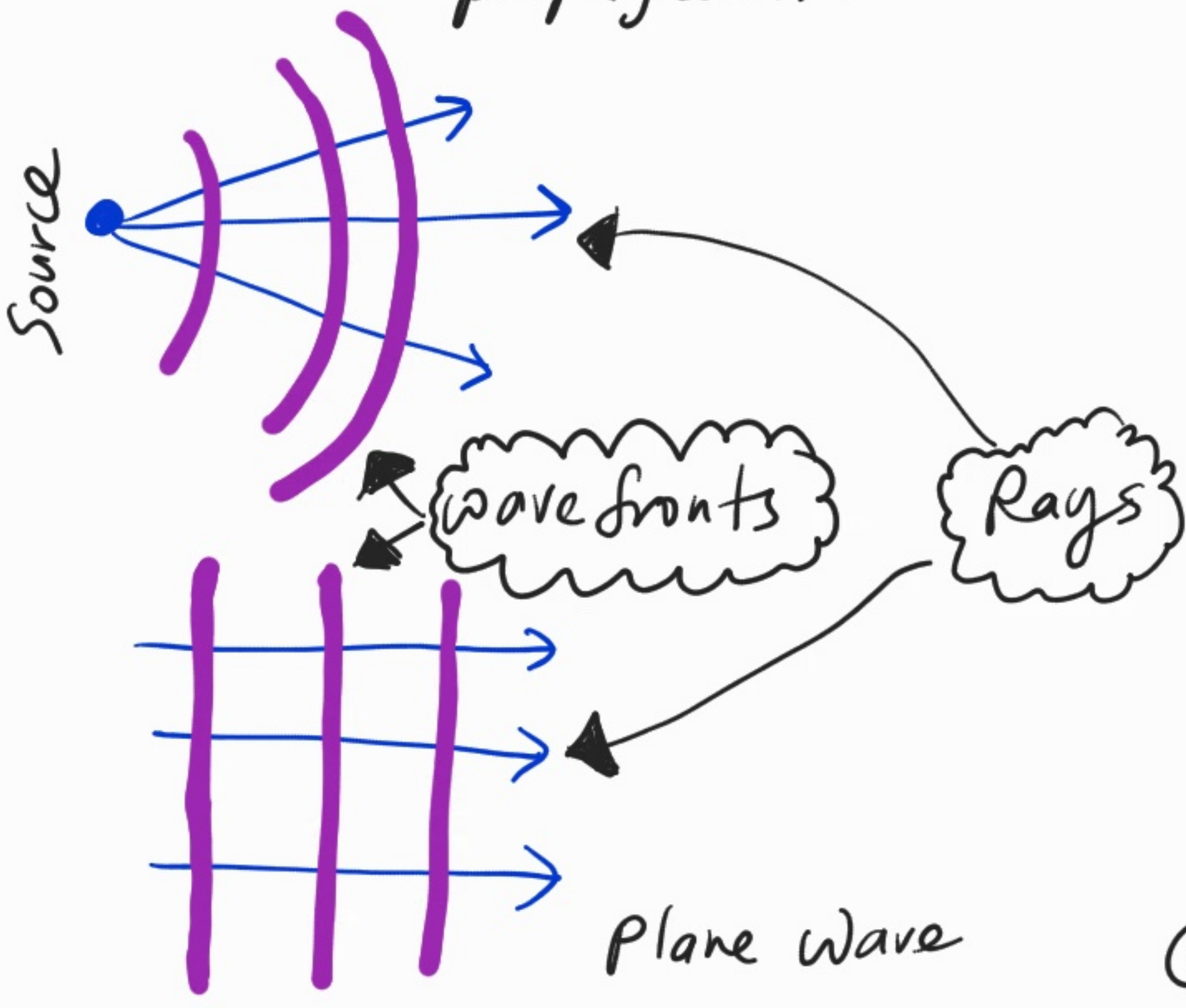
Wavefronts & Rays

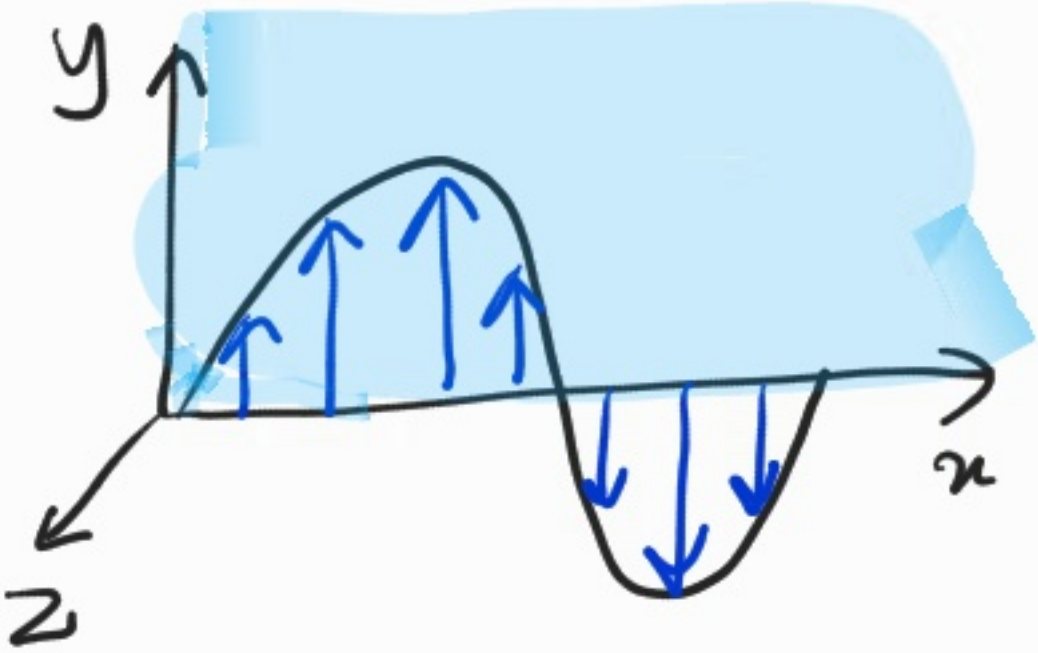
Wavefront → used to describe wave propagation

Ray → used to describe the the direction of propagation.

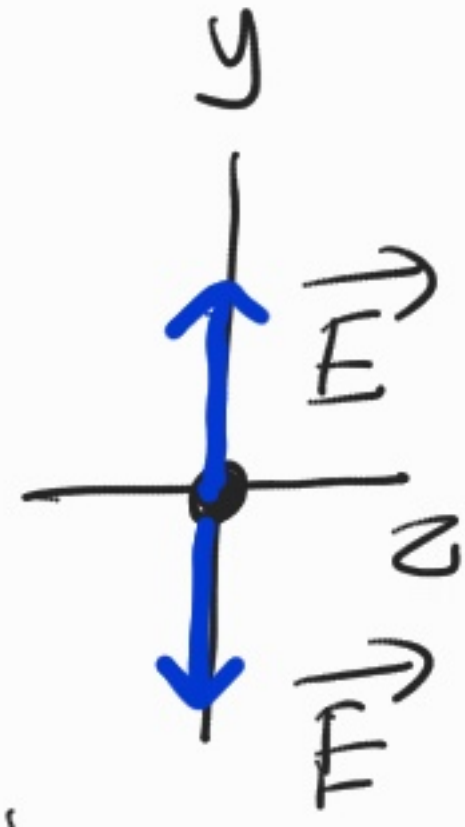


Polarization of Light

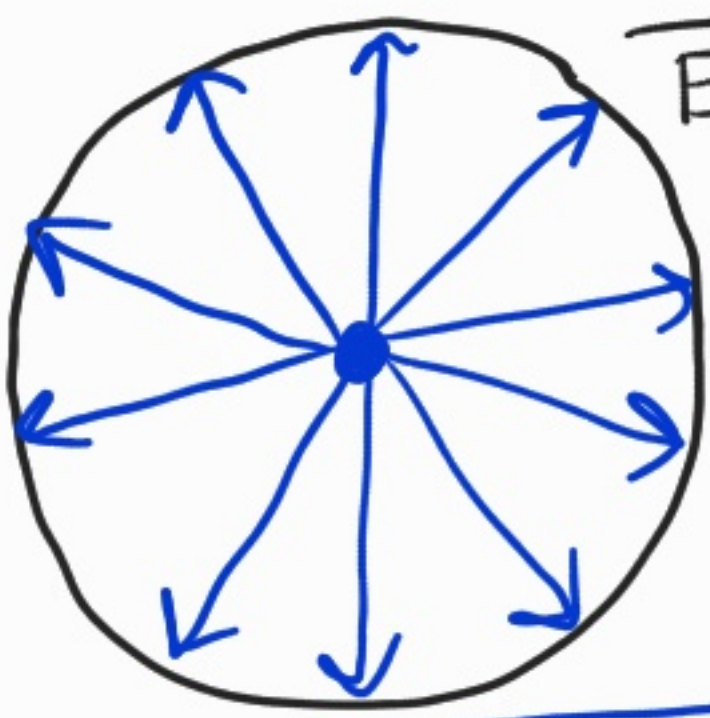
Electric field oscillates vertically



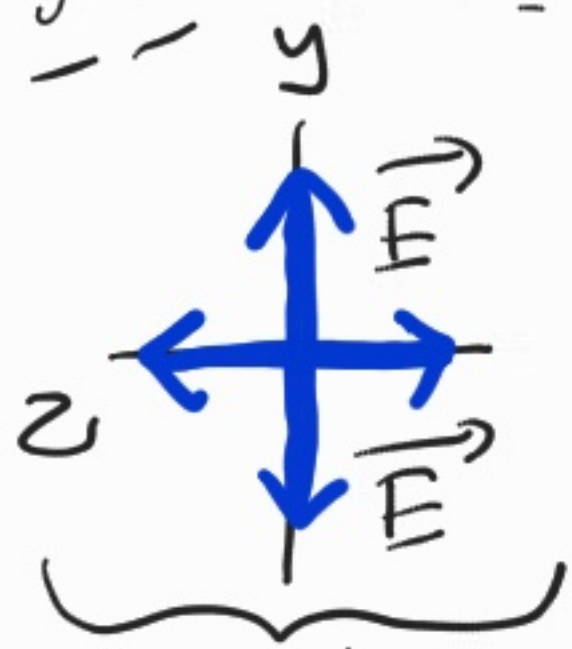
Polarized Light



symbol for vertically polarized light



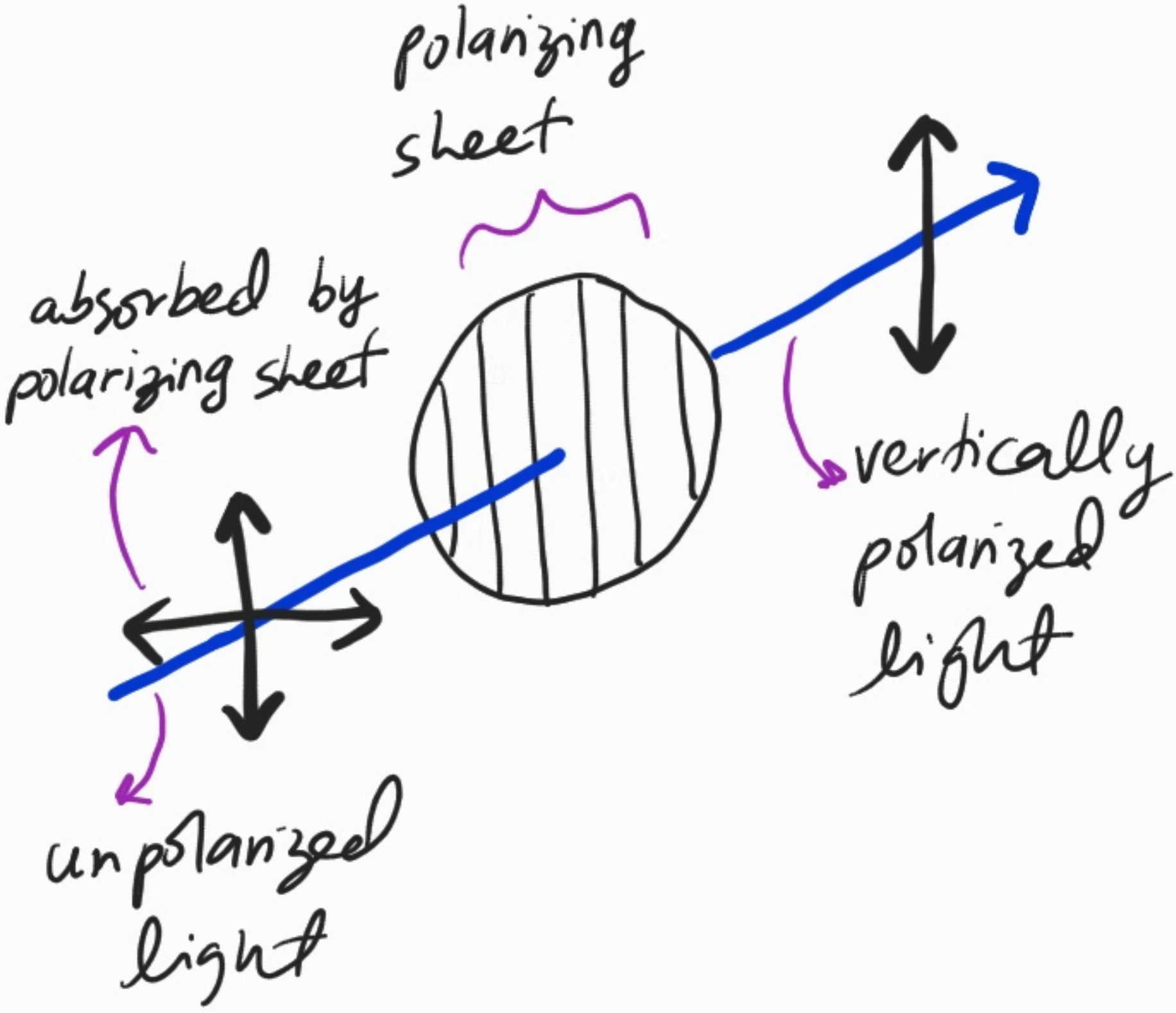
Unpolarized Light



symbol for unpolarized light

Polarizing Sheets

⇒ used to transform unpolarized light (visible) to polarized light.



Polarizing Sheets

- For polarized light, 1/2 of the light is absorbed by the polarizing sheet.
- Intensity of transmitted polarized light:

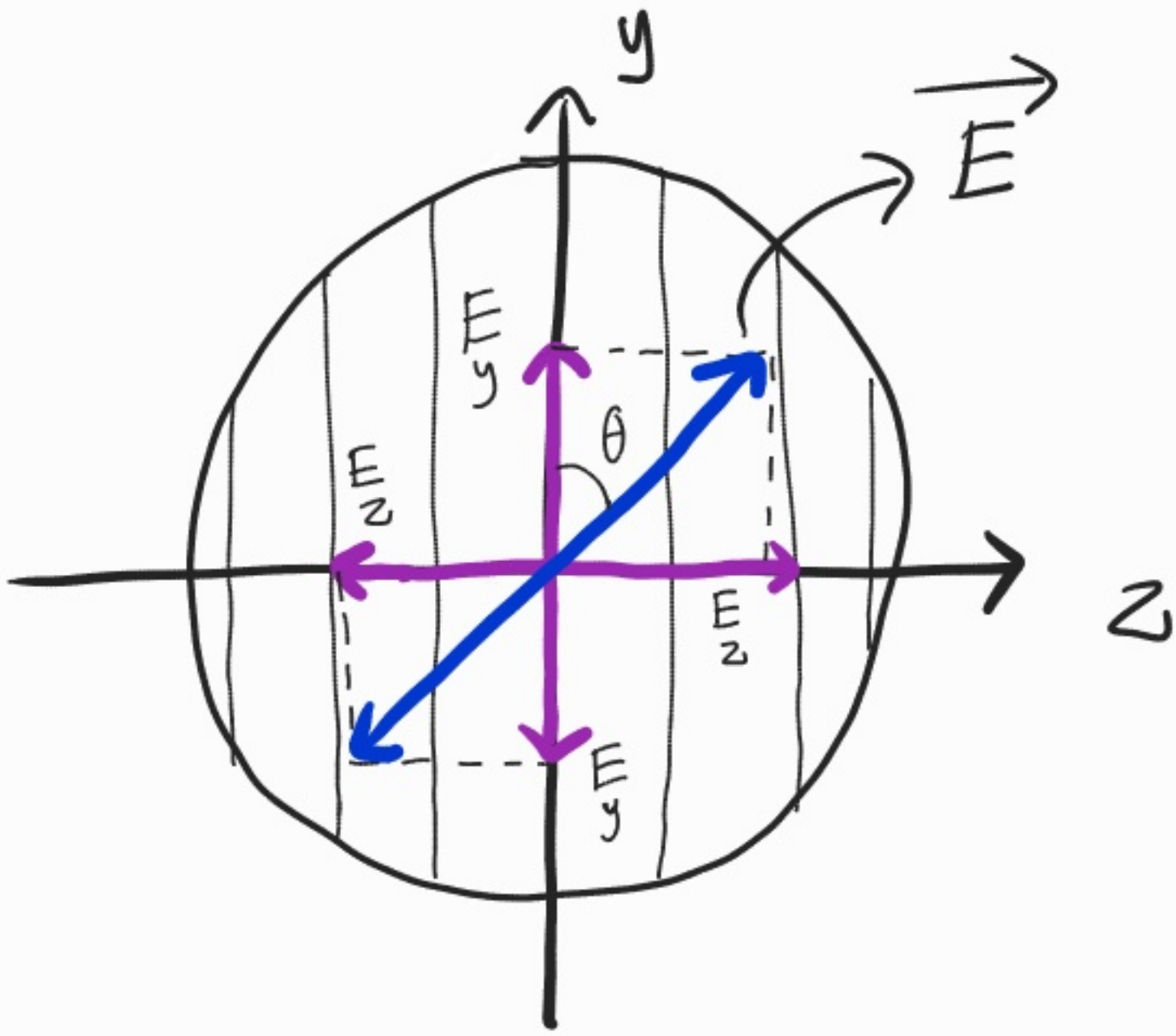
$$I = \frac{1}{2} I_0$$

One-Half Rule

I → Polarized light

I_0 → original unpolarized light

Polarizing Sheets



⇒ only the y-component of the electric field passes trough the polarizer.

$$E_y = E \cos\theta$$

Polarizing Sheets

⇒ Intensity of unpolarized light:

$$I_0 \propto E^2$$

⇒ Intensity of polarized light:

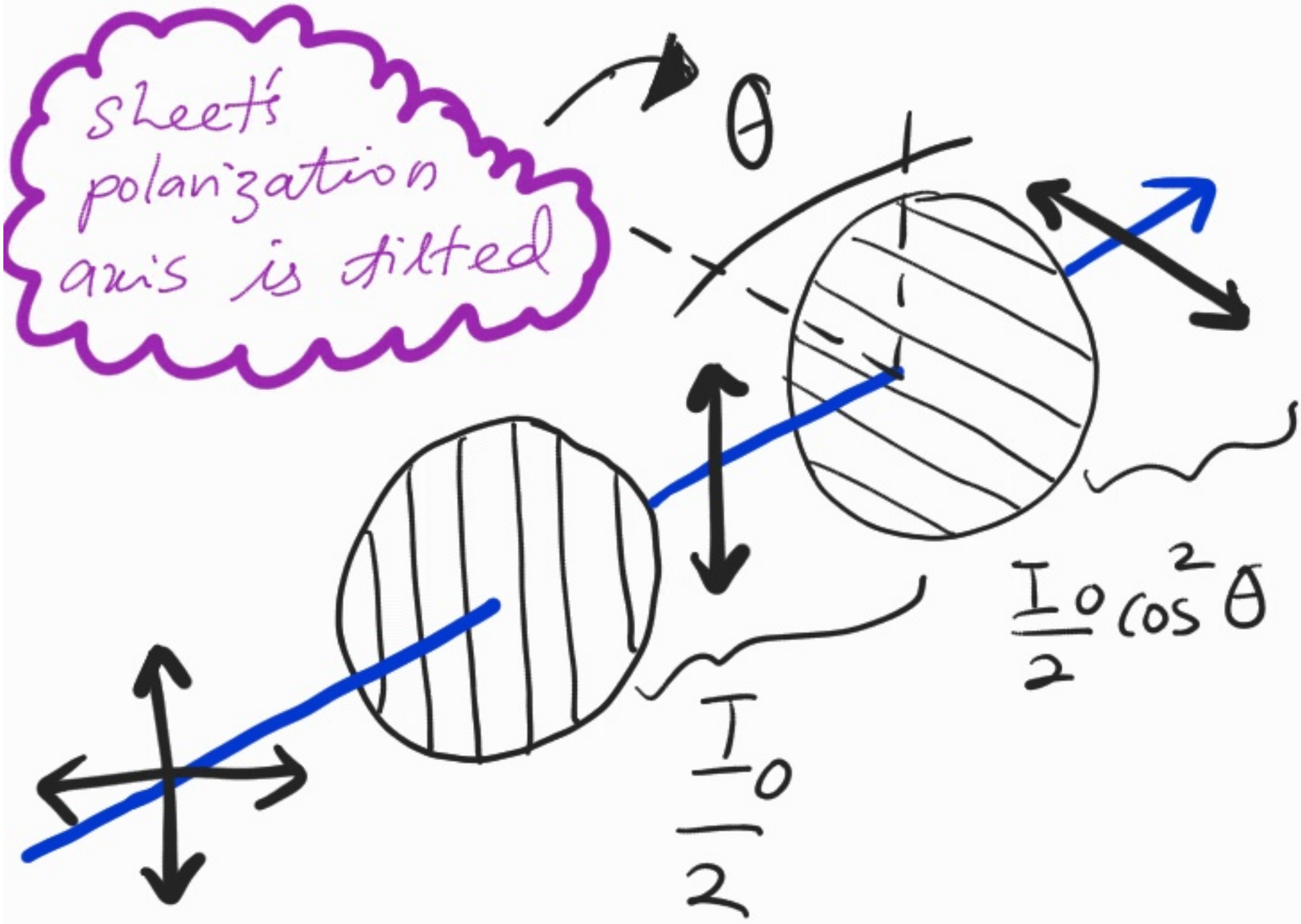
$$I \propto E^2 \cos^2\theta$$

COSINE SQUARED RULE

$$\therefore \frac{I}{I_0} = \cos^2\theta \rightarrow I = I_0 \cos^2\theta$$

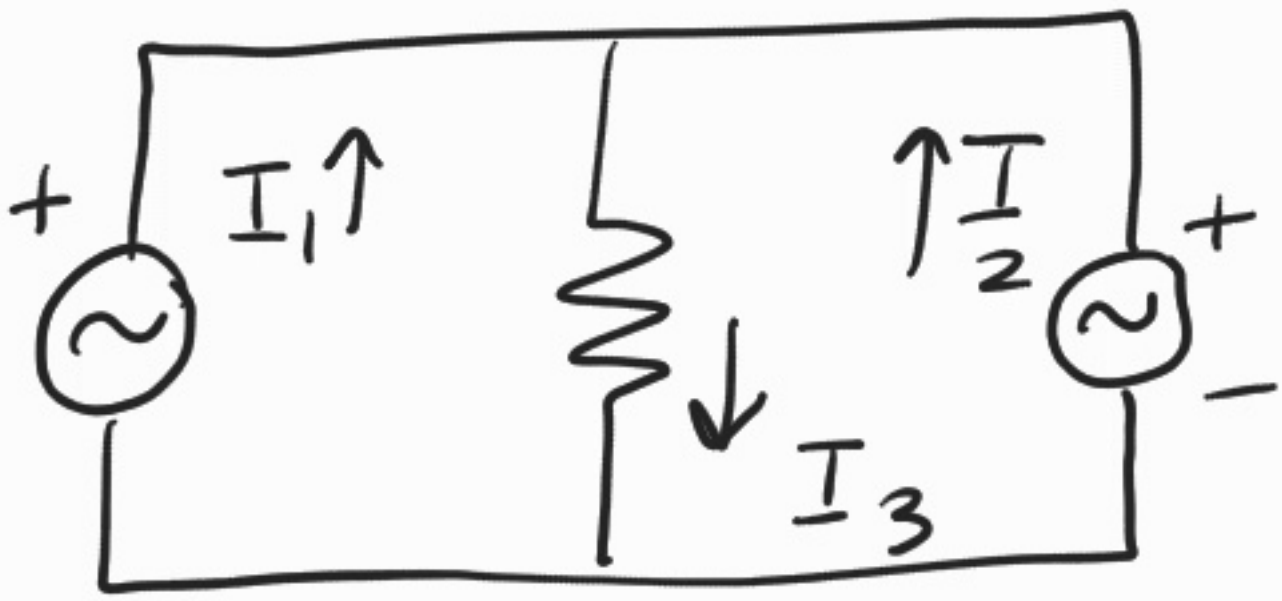
Polarizing Sheets

- Two polarizing sheets.



Practice Problem 1 AC Superposition

Consider the following AC circuit:



Find the complex amplitude & the phase constant of $I_3(t)$ if:

$$I_1(t) = 10.5 \sin(\omega t + \pi/4)\ A$$

$$I_2(t) = 5.8 \cos(\omega t - \pi/4)\ A$$

Strategy.

1, Phase shift I_1 to match the phase of I_2

2, Represent both currents in their phasor form.

3, Sum both phasor currents to obtain superposition:

$$\tilde{I}_3 = \tilde{I}_1 + \tilde{I}_2$$

4, Determine amplitude & phase constant.

Solution:

$$I_1(t) = 10.5 \sin\left(\omega t + \frac{\pi}{4}\right)$$

$$= 10.5 \cos\left(\omega t + \frac{\pi}{4} - \pi/2\right)$$

$$= 10.5 \cos(\omega t - \pi/4)$$

$$\frac{I}{2}(t) = 5.8 \cos(\omega t - \pi/4)$$

Phasor Representation:

$$e^{i\theta} = \cos\theta + i \sin\theta$$

$$\tilde{x}(t) = x_m e^{i(\omega t + \phi)}$$

$$x(t) = Re(\tilde{x}(t))$$

$$= x_m \cos(\omega t + \phi)$$

Solution. (Con't)

$$I(t) = 10.5 \cos(\omega t - \pi/4)$$

$$I_2(t) = 5.8 \cos(\omega t - \pi/4)$$

$$\tilde{I}_1(t) = 10.5\, e^{j(\omega t - \pi/4)}$$

$$\tilde{I}_2(t) = 5.8\, e^{j(\omega t - \pi/4)}$$

$$\tilde{I}_3(t) = \tilde{I}_1(t) + \tilde{I}_2(t)$$

$$= 16.3\, e^{-j\pi/4} e^{j\omega t}$$

$$\tilde{I}_{3,m}(t) = 16.3\, e^{-j\pi/4}$$

complex amplitude

phase constant: $-\pi/4$ #

of the superposition

Practice Problem 2

Determine the complex amplitude & the phase constant of I_3 in Problem 1 if the current $I_2(t)$ has the following form:

$$I_2(t) = -5.8\cos\left(\omega t + \frac{3\pi}{4}\right)$$

Strategy:

1/ Use the following identity to transform $\tilde{I}_2(t)$:

$$e^{-2i\pi} = -1$$

2/ Check if $\tilde{I}_2(t)$ is in phase with $\tilde{I}_1(t)$, then perform sum in phasor form to obtain $\tilde{I}_3(t)$ [superposition].

Solution:

$$I_2(t) = -5.8\cos\left(\omega t + \frac{3\pi}{4}\right)$$

$$-1 = e^{-i\pi}$$

$$\tilde{I}_2(t) = -5.8\, e^{i\left(\omega t + \frac{3\pi}{4}\right)}$$

$$\tilde{I}_2(t) = 5.8\, e^{i\omega t}\, e^{i3\pi/4}\, e^{-i\pi}$$

$$= 5.8\, e^{-i\pi/4}\, e^{i\omega t}$$

$$\tilde{I}_1(t) = 10.5\, e^{i(\omega t - \pi/4)}$$

$$= 10.5\, e^{-i\pi/4}\, e^{i\omega t}$$

$$\tilde{I}_3(t) = \tilde{I}_1(t) + \tilde{I}_2(t)$$

$$= 16.3\, e^{-i\pi/4}\, e^{i\omega t}$$

$$\underbrace{\tilde{I}_{3,m}}$$

#